The proposed 16-chapter outline

Using Data to Drive Business Performance

Research and proposal review · October 2026

Things that shaped the structure of this courseware

Seven years in the classroom	I have taught the introductory MIS course at the University at Albany for seven years, to more than 4,000 students. The chapter labs and quizzes were developed and refined across those semesters.
More than ten years of editing	Before I became a professor, I spent over ten years as a magazine editor. That shaped the writing: clear, visual and organized around the reader, at a level business undergraduates can follow.
Extensive market research	I reviewed and analyzed the main textbooks used in these courses, surveyed what universities across the country teach, and compared 63 syllabi collected from 31 institutions. The outline draws directly on that work. See the research below.
Full control and quick iteration	Because I built the platform, tools, examples and AI guidance can change each semester instead of each edition. After the summer test and the current deployment with 500 students, I collect live feedback from students and revise continuously.

The course arc

The proposed 16-chapter edition separates administration, analytics, visualization and Power BI, and gives the capstone its own chapter. Semester workload and the two course routes remain open to publisher review.

Data → Tables → Relationships → Queries → Analytics → Decisions

The following outline is based on the detailed research presented below.

The book at a glance

A preface, **16 chapters in five parts** , a midterm review week, a closing chapter on careers and seven appendices. It offers a proposed route for a 14- to 16-week semester, subject to course-level workload review. Each part below opens with its main goal, and each chapter lists its sections, its guided build and lab, and the research it is based on.

PREF Preface: Why This Book, and How It Works

Why it was written and how to use it.

• P.1 Why I Wrote This Book

• P.2 How This Book Is Different

• P.3 How a Chapter Works

• P.4 The Tools, and Why They Were Chosen

• P.5 Two Routes Through the Book

• P.6 How AI Is Used in This Book

Part I · Chapters 1 to 3

From Data to Decisions

Main goal: Connect data to business performance, classify data correctly and organize information.

1 From Data to Business Performance

How does a business turn everyday activity into better results?

Let's Build: Trace one KPI back to the records that produce it.

Interactive lab: A KPI audit of a small business: three KPIs, the data each needs, and who decides with it.

• 1.1 What a Business Is and How It Creates Value

• 1.2 Business Performance, Goals and KPIs

• 1.3 From Data to Wisdom: The DIKW Pyramid

• 1.4 Information Behavior: How People Seek and Use Information

• 1.5 Information Systems and Information Technology

- 1.6 The R.E.A.D. Framework: Recognize, Extract, Analyze, Decide
- 1.7 AI in the Data-to-Decision Chain
- 1.8 The Running Cases and the Semester Project

Based on: None of the four leading texts opens with business performance and KPIs. Business-systems framing appears in Buffalo MGS 351, Syracuse IST 195 and UNLV MIS 766.

2 Understanding Data

What kind of data is this, and what can I legitimately do with it?

Let's Build: Profile a dataset column by column: type, scale and allowed operations.

Interactive lab: Data Type Detective: classify and repair a messy business file, then draw one valid insight per column type.

- 2.1 Data and Information
- 2.2 Digital and Analog Data; Bits, Bytes and Tokens
- 2.3 Data Types: Numbers, Text, Dates, Booleans and Files
- 2.4 Qualitative and Quantitative Data
- 2.5 Levels of Measurement: Nominal, Ordinal, Interval, Ratio
- 2.6 Structured, Semi-Structured and Unstructured Data: CSV and JSON
- 2.7 What Makes a Good Table
- 2.8 Data Quality
- 2.9 From Data to Insight: What Each Type Allows

Based on: The compared outlines do not give data literacy and measurement scales their own opening chapter. Data preparation appears in NYU TECH-UB 26 and Temple MIS 2502.

3 Organizing Data and Information

How do we make data findable, reusable and trustworthy?

Let's Build: A data dictionary and a folder-and-tag scheme for the running case.

Interactive lab: Reorganize and document a shared-drive mess.

- 3.1 Why Organization Matters
- 3.2 Files, Folders, Naming and Versions
- 3.3 Hierarchies, Tags and Facets
- 3.4 Metadata, with YAML as an Example
- 3.5 Taxonomies and Controlled Vocabularies
- 3.6 Data Dictionaries
- 3.7 Spreadsheets as Tables, and Where They Break
- 3.8 From Organized Files to Databases

Based on: Added by design from information science, so students see why databases exist before they build one. Not a standalone topic in the syllabi reviewed.

Part II · Chapters 4 to 6

Databases and SQL

Main goal: Read a relational database and answer business questions with SQL.

4 Databases and the Relational Model

Why do organizations keep data in related tables instead of files?

Let's Build: Explore a supplied case database in Supabase Studio and trace a relationship.

Interactive lab: Read an unfamiliar schema and predict the answers to five business questions before querying.

- 4.1 What a Database Is and What a DBMS Does
- 4.2 Local and Cloud Databases: SQLite, PostgreSQL, Supabase, MySQL, Access
- 4.3 Tables, Rows, Columns and Data Types
- 4.4 Primary, Foreign and Composite Keys
- 4.5 Relationships: One-to-One, One-to-Many, Many-to-Many
- 4.6 Schemas, Integrity Rules and Constraints
- 4.7 Database Objects: Tables, Queries, Views, Forms and Reports

Based on: Design and relational foundations appear in every business-database course description reviewed, including Baruch CIS 3400, George Mason MIS 310 and SUNY Canton CITA 215.

5 SQL Foundations

How do I ask one table a precise question?

Let's Build: Answer ten business questions on the running case.

Interactive lab: The same skills on the second case, with verification tasks.

- 5.1 SELECT and FROM
- 5.2 Filtering with WHERE, LIKE, IN and BETWEEN
- 5.3 Sorting, DISTINCT and LIMIT
- 5.4 Working with NULL and Dates
- 5.5 Aggregate Functions
- 5.6 Creating Tables and Inserting Data
- 5.7 Safe UPDATE and DELETE
- 5.8 Verify an AI-Written Query

Based on: SQL appears in 10 of 13 catalogue benchmark courses (77%) and in 52.9% of analyst job postings. Early SQL in Rutgers 33:136:470, Northeastern CS 3200 and UT Dallas ITSS 4300.

6 SQL for Business Analysis

How do I combine tables into trustworthy business measures?

Let's Build: Build a small KPI report from joined tables.

Interactive lab: Find and fix a double-counted measure.

- 6.1 INNER and LEFT Joins
- 6.2 Joining Three or More Tables
- 6.3 GROUP BY and HAVING
- 6.4 CASE Expressions for Business Categories

- 6.5 Calculated Measures: Rates, Averages and Shares
- 6.6 Grain and the Double-Count Problem
- 6.7 Checking Results by Hand

> Based on: Joins and aggregation are where business reports most often go wrong. Taught at Rutgers, Temple, UT Austin and UT Dallas.

MID Integration Studio A (midterm)

An unnumbered review week between Parts II and III.

Cumulative review of Chapters 1 to 6, a practice exam, and **Project Checkpoint 1** : proposal and data plan.

> Based on: most of the courses reviewed run staged, multi-phase projects, so the project has graded checkpoints through the semester.

Part III · Chapters 7 to 11

Designing and Running Databases

Main goal: Design, normalize, build, test and secure a database of their own.

7 Database Design: From Requirements to ER Diagrams

How do business rules become a database design?

Let's Build: Draw the ERD for the running case from its business rules.

Interactive lab: Design an ERD for a new business process.

- 7.1 The Systems Development Life Cycle
- 7.2 Waterfall and Agile
- 7.3 Requirements and Business Rules
- 7.4 Entities, Attributes and Relationships
- 7.5 Cardinality, Optionality and Crow's Foot Notation
- 7.6 Associative Entities and Many-to-Many Relationships
- 7.7 Drawing ERDs with Lucidchart, diagrams.net or Mermaid

Based on: Design and modeling appear in every business-database course reviewed. ER modeling comes before normalization, as in Coronel, Kroenke and Hoffer.

8 Normalization

How do we design out errors before they happen?

Let's Build: Normalize a flat spreadsheet to third normal form.

Interactive lab: Normalize a real business form and prove no data was lost.

- 8.1 Update, Insert and Delete Anomalies
- 8.2 Functional Dependencies
- 8.3 First Normal Form
- 8.4 Second Normal Form
- 8.5 Third Normal Form
- 8.6 Decomposing and Rejoining Tables
- 8.7 When to Denormalize

Based on: Normalization is core in every competing text and in Baruch, Rutgers, George Mason, Georgia Tech and SUNY Farmingdale BCS 260.

9 Building and Testing a Database

How do we turn a design into a working, trustworthy database?

Let's Build: Build and load the running-case database in Supabase.

Interactive lab: Build the lab case from its ERD and run an integrity test suite.

- 9.1 Data Definition Language: CREATE, ALTER, DROP
- 9.2 Constraints
- 9.3 Loading Data from CSV
- 9.4 Staging and Cleaning
- 9.5 Integrity Tests

• 9.6 Data Entry Workflows and Forms

Based on: Hands-on building appears in 10 of 13 catalogue benchmark courses (77%). Implementation projects at Rutgers, Baruch and Boston University MET CS 669.

10 Advanced SQL

How do analysts write queries that are reusable and easy to check?

Let's Build: A reusable analysis layer of views and CTEs.

Interactive lab: A monthly performance report with rankings and period-over-period change.

• 10.1 Subqueries

• 10.2 Common Table Expressions

• 10.3 Views

• 10.4 Conditional Aggregation

• 10.5 Window Functions

• 10.6 Set Operations

• 10.7 Functions and Triggers

• 10.8 Reading a Query Plan with EXPLAIN

Based on: Analytical SQL with CTEs and window functions appears in current syllabi at NYU Stern and UT Dallas.

11 Database Administration, Security and Reliability

How do we keep business data safe, correct and available?

Let's Build: Add roles, row-level security and an audit trigger to the running case.

Interactive lab: A security and recovery plan for the lab case. Project Checkpoint 2: working database.

• 11.1 Roles, Users and Least Privilege

• 11.2 Row-Level Security

• 11.3 Transactions and ACID

- 11.4 Concurrency
- 11.5 Backup and Recovery
- 11.6 Indexes and Performance
- 11.7 Governance, Privacy and Compliance
- 11.8 Audit Trails

Based on: Administration, security or transactions appear in 6 of 13 catalogue benchmark courses (46%), including Rutgers, UT Austin, George Mason and SUNY Canton.

Part IV · Chapters 12 to 15

From Data to Insight

Main goal: Analyze results, visualize them, report in Power BI and work with warehouses and AI.

12 Business Analytics and Decision-Making

What does the data actually tell us, and how sure are we?

Let's Build: A SQL analysis that answers one management question.

Interactive lab: An analytical memo for the lab case.

- 12.1 Four Kinds of Analytics
- 12.2 Framing an Analytical Question
- 12.3 Distributions, Segments and Fair Comparisons
- 12.4 Correlation and Causation
- 12.5 Uncertainty
- 12.6 From Measure to Decision
- 12.7 The Decision Memo

Based on: Closes the loop from query to decision, which few database texts do. Business analysis tasks in Temple MIS 2502 and Georgia MIST 4610.

13 Reporting and Data Visualization

How do we show results so people decide well?

Let's Build: Redesign a misleading dashboard.

Interactive lab: A one-page dashboard specification.

- 13.1 Reports and Dashboards
- 13.2 Choosing the Right Chart
- 13.3 Honest and Misleading Visuals
- 13.4 Accessible Design
- 13.5 Designing for Decision-Makers
- 13.6 The Tableau Route

Based on: Only 4 of 13 catalogue benchmark courses (31%) include BI or analytics. Visualization appears in Georgia MIST 4610 and UT Austin MIS 325.

14 Power BI

How do we turn a database into an interactive business report?

Let's Build: A Power BI report on the running case.

Interactive lab: A Power BI report on the lab case, validated against SQL.

- 14.1 Connecting to a Database
- 14.2 Shaping Data with Power Query
- 14.3 The Data Model and Relationships
- 14.4 Measures, Calculated Columns and DAX Basics
- 14.5 Building an Interactive Report
- 14.6 Validating Every Number Against SQL

Based on: Power BI is requested in 29% of analyst job postings and appears in none of the syllabi reviewed, so it is added by design.

15 Big Data, Data Warehouses and AI

How do organizations work with data at scale, and where does AI fit?

Let's Build: A small star schema from the running case.

Interactive lab: Evaluate three AI answers about the lab data: accept, revise or reject.

- 15.1 OLTP and OLAP
- 15.2 Warehouses, Marts, Lakes and Lakehouses
- 15.3 Star Schemas: Facts and Dimensions
- 15.4 ETL, ELT, APIs and JSON
- 15.5 Big Data and NoSQL
- 15.6 AI and Data: Text-to-SQL, Embeddings and Retrieval
- 15.7 Verifying AI Output

> Based on: The 13-course catalogue benchmark explicitly names AI / ML in one description and cloud / APIs / ETL in none. Warehousing in Baruch CIS 4400 and graduate AI topics in UT Dallas inform extensions; these do not establish universal requirements.

Part V · Chapter 16 and Closing

Putting It to Work

Main goal: Deliver a capstone decision package and plan a path into data careers.

16 Capstone: From Data to Decisions

Can you take a real business problem all the way to a defensible decision?

Project Checkpoint 3: the final project, with a working database, analysis, dashboard and decision memo, and an individual demonstration.

- 16.1 Scoping the Problem and the KPI
- 16.2 From Data Dictionary to ERD
- 16.3 Building the Database
- 16.4 Analysis and a Validated Dashboard
- 16.5 The Decision Memo

• 16.6 Demonstration and Reflection

Based on: Staged projects at Rutgers, Boston University and Georgia. The individual demonstration answers the AI-policy gap found in the syllabi.

END Closing: Where Data Can Take You

Where do these skills lead?

• C.1 The Data-to-Decision Journey on One Page

• C.2 Careers in Data: Analyst, BI Analyst, Database Administrator, Data Architect, Data Engineer

• C.3 Data Skills in Every Business Major

• C.4 Building a Portfolio

Based on: The labor-market figures in the market research: data scientist jobs growing 35% and database architect jobs 9% through 2035.

Appendices

A

Supabase and the In-Book SQL Editor

Setting up a Supabase project, using Supabase Studio, and running PostgreSQL inside the reader.

B

MySQL Route

Tested schemas, seed data and solutions for every SQL chapter, for courses built on MySQL.

C

Microsoft Access Route

Tables, relationships, visual queries, forms and reports on the same cases.

D

SQL Reference and Dialect Guide

PostgreSQL, MySQL, SQL Server and SQLite side by side.

E

Diagramming with Lucidchart

ER diagrams in Lucidchart, with diagrams.net and Mermaid as free alternatives.

F

Tableau Route

The visualization chapter's activities in Tableau.

G

Aa

Terms Treasury

The glossary of every key term in the book.

How students progress

• **Understand the data** Chapters 1 to 3: what data is, how it measures performance, and how to organize it.

• **Use a database** Chapters 4 to 6: query a supplied database before designing one.

• **Build one** Chapters 7 to 11: design, normalize, build and secure their own database.

• **Decide and defend** Chapters 12 to 16: turn results into dashboards and a decision memo they can explain and defend.

Two ways to use the book

Both routes come from the syllabus research: business-core courses stop at design and reporting, while MIS courses need the full database sequence.

Business-core data course	Chapters 1 to 8, then 12 to 14.
MIS database course	Chapters 1 and 4 to 16, with Chapters 2 and 3 as assigned reading.

The syllabus platform audit distinguishes documented practical use, a named option, and a software-list mention. Counts use 54 selected course families and overlap across tools. Official

documentation then informs the proposed PGlite, Supabase, and Power BI pathway. [S2] [T1] [T2] [T3]

Practical use means the reviewed text assigns or demonstrates work with the tool; an option does not establish that a student selected it. Twenty course families lack sufficiently explicit practical / listed platform evidence. No adoption of Supabase or PGlite was established in this archive.

Main conclusion

MySQL and Access have substantial documented footprints in the reviewed courses. Keep meaningful compatibility routes while using one principal SQL foundation. The choice of PostgreSQL is a design recommendation, rather than the dominant platform in this sample.

The proposed technology pathway

Recommendation: PostgreSQL as the SQL foundation, PGlite for immediate practice in the browser, and Supabase for the hosted project. Power BI provides the main reporting studio; Tableau is an alternate route. Lucidchart supports modeling, with a supported alternative. Access and MySQL receive substantive companion coverage.

Layer	Recommended role	Condition for adoption
SQL	Main skill and reasoning language.	Portable core examples with PostgreSQL-specific syntax labeled.
PGlite	Immediate embedded practice.	Implement and test persistence, reset, accessibility and known-answer checks; proposed, not yet verified here.
Supabase	Hosted project and operational controls.	Classroom onboarding, export/restore, permission tests and a supported exit route; no guaranteed free service claim.
Power BI	Principal BI studio in Chapter 14.	Windows authoring access or a supported institutional virtual lab.
Tableau	Equivalent alternate visualization studio.	Same data, measures and rubric; avoid two required BI tools in the standard route.
Lucidchart	Diagramming example.	Provide a supported alternative; no presumed free-tier entitlement.

Layer	Recommended role	Condition for adoption
Access / MySQL	Substantive compatibility routes.	Test equivalent core examples and instructor setup; not every student must learn both.

PGlite executes locally in the browser; Supabase is hosted PostgreSQL. Export / import must be taught explicitly. Power BI Desktop requires Windows or an institutional virtual lab. “Browser-first SQL practice with a hosted project” accurately describes the intended path; integration, access, licensing, and maintenance still need implementation decisions.

Companions and appendices Platform setup, alternatives, reference, and instructor resources

Companion	Purpose
PostgreSQL / PGlite / Supabase	Setup, workspace reset/persistence, portable SQL, explicit transfer, exports, restore and access tests.
MySQL / MySQL Workbench	A tested alternate route with matching data, query results and dialect differences. An appendix-length syntax list alone is insufficient.
Microsoft Access	Tables, relationships, visual queries, forms, reports and an equivalent small application.
Lucidchart and alternatives	Crow’s Foot modeling, exports and review; diagrams.net/ERDPlus or Mermaid alternatives avoid a mandatory subscription.
Power BI and Tableau	A complete Power BI studio plus an equivalent Tableau reporting route; Mac access planning is explicit.
SQL reference and advanced database topics	Portable core syntax, dialect flags, triggers/procedures, deeper windows and normal forms.
Capstone, reviews and instructor resources	Milestones, tests, rubrics, solution files, slides, question banks and course-route maps.

Running cases

Every Let's Build uses one guided case and every lab uses a second case, so students learn a skill once and then transfer it. The case choice is one of the open questions.

Guided case

Grading database

Courses, assignments and grades offer a familiar setting, although policy complexity still needs careful scoping.

What to test: Keep as the guided case with a small, stable schema.

Lab case

PetVax veterinary clinic

Customers, pets, visits, services and payments give real relationships and operating KPIs.

What to test: Whether students relate to the setting.

Alternative lab case

Small retail business

Orders, products, customers and revenue suit the analytics, Power BI and star-schema chapters.

What to test: Whether the extra entities overload an introductory lab.

Eight recurring chapter components

• **Introduction.** The business question and the chapter objectives.

• **Core concepts.** The main chapter, in numbered sections with worked examples.

• **Terms Treasury.** The key terms, defined in context.

• **Review questions.** Recall and reflection.

• **Reading Assessment Test.** An auto-graded quiz, tied to the book's own databases.

• **Let's Build.** Guided, hands-on practice on the running case.

• **Interactive lab.** Independent practice on a second case, with hints and verification tasks.

• **Key takeaways.** A summary and a bridge to the next chapter.

• **Semester project.** Three graded checkpoints across the book, ending in a capstone decision memo.

Assessment that makes reasoning visible

Students predict and define before execution, test keys and totals during execution, and interpret the result afterward. Reading checks, guided practice, independent labs, cumulative reviews, and the final project serve distinct purposes.

Assessment	Purpose	Proposed evidence
Reading assessment tests	Check readiness and conceptual distinctions.	Short scenario, output-prediction and misconception items; separate formative feedback from graded evidence.
Let’s Build	Guided skill development.	Worked reasoning, expected results and checkpoints.
Interactive chapter labs	Independent transfer.	SQL, numeric answers, diagrams, files and short interpretations, appropriate to the chapter.
Cumulative tests / reviews	Integrate ideas across chapters.	Unfamiliar schema or changed business rule; explain and repair.
Final project	Demonstrate a complete usable solution.	Rebuildable database, validation log, report, decision memo and individual demonstration.

Assessment principle

Use known-result SQL and integrity tests where they fit; use human review for modeling, explanations, and business judgment. Evaluate unfamiliar variations and require a short individual explanation. AI-assisted feedback is a development option that needs validation.

One problem. Three checkpoints. A defensible recommendation.

• **Question and requirements.** A scoped business problem, stakeholder, measure, source plan and acceptance criteria. Start in the opening chapters.

• **Design and working data.** An ERD, normalization rationale, implemented schema, synthetic data, import/reject log and integrity checks.

• **Evidence and action.** Tested SQL, reconciled reporting, essential access/recovery controls, a decision memo and an individual change task.

Proposed rubric	Weight
Problem framing and requirements	10%
Data model and normalization rationale	20%
Implementation and data quality	20%

Proposed rubric	Weight
SQL correctness and verification	20%
Reporting and business interpretation	15%
Reproducibility, controls and individual explanation	15%

Rubric weights are a proposal for faculty review, not a change to the live course. A compact submission includes schema.sql, seed data, queries.sql, an ERD, validation evidence, README and a short decision memo. A live link alone is insufficient because its data may change after submission.

References

Market, employment and curriculum references

References

• National Center for Education Statistics. (2023). Digest of Education Statistics, Table 318.20: Degrees conferred by field of study, 2021-22. Link

• Fustos, J., et al. (2017). Do students get enough information systems education in the business core? Study of 61 business schools. Link

• Data USA. (2026). Management information systems and services (IPEDS completions, 2024). Link

• Hwang, D., Ma, Z., & Wang, M. (2015). The information systems core: A study from the perspective of IS core curricula in the U.S. Information Systems Education Journal, 13(6), 27-34. Link

• Zheng, Y., Hameed, T., Lavoie, R., & Sendall, P. (2021). An overview of current business analytics programs across US AACSB schools. Issues in Information Systems, 22(2), 306-317. Link

• Yang, S. C. (2023). The master's program in information systems: A survey of core curricula in AACSB-accredited business schools in the United States. Journal of Information Systems Education, 34(4), 430-440. Link

• Data USA. (2026). Accounting (IPEDS completions, 2024). Link

• AACSB International. (2026, September). Accreditation announcement, September 2026. Link

• 365 Data Science. (2026). The data analyst job market in 2026: Research on 1,000 job postings. Link

• U.S. Bureau of Labor Statistics. (2026). Occupational Outlook Handbook: Data scientists. Link

• U.S. Bureau of Labor Statistics. (2026). Occupational Outlook Handbook: Database administrators and architects. Link

• World Economic Forum. (2025). The Future of Jobs Report 2025. Link

• Cengage. Concepts of Database Management, 10th edition (2021). Link

Manuscript, textbook, platform and original-cover references

[S1]

Collected syllabus archive and manifest ↗

63 documents; 31 institutions; archived October 1, 2026. Mixed course levels and dates.

[S2]

Database-platform evidence model and report ↗

October 17, 2026 audit. Role-coded platform evidence, 54 selected course families; counts recalculated for this package.

[S3]

Curriculum audit ↗

October 1, 2026 document-level topic summaries and original-source references; used as a discovery aid and checked against archived course text.

[S4]

Textbook adoption and contents audit ↗

October 10, 2026 report: 60 named books/course texts. Historical school-level mentions, not market shares. Broader contents catalogue retained as prior research.

[S5]

Stabilized 14-chapter proposal ↗

Earlier synthesis, stored in the outline-comparison folder. This package proposes a refinement; it does not replace the manuscript outline.

[S6]

Current book repository ↗

Read-only inspection of main, commit 92d7a6ea04b96b67ef983aa22d054ab0bb9ad046; current chapter activation metadata checked.

[S7]

Inactive book-2027 candidate ↗

16-chapter migration candidate with separate core/extension routes. Its README explicitly states that it is inactive and is not published by the reader.

[S8]

Dima Publishing design and book description ↗

Publisher design reference and earlier public description. Cover for this revision comes from the current book website instead; see C1.

[S9]

Brightlab evaluation design document ↗

Prototype description, not validation of fairness, reliability, costs or deployability. Contradictory partial-credit claims require resolution.

[P1]

Cengage: Database Systems, 14th edition ↗

Coronel and Morris. Uploaded copy supplies the historical chapter sequence; official product page documents interactive MySQL labs and other MindTap resources.

[P2]

Cengage: Database Systems, 15th edition ↗

Coronel and Morris. Official current listing checked October 25, 2026; detailed sequence is retained from the earlier contents audit. Recent edition moves SQL ahead of the formal design sequence.

[P3]

Cengage: Concepts of Database Management, 10th edition ↗

Friedrichsen, Ruffolo, Monk, Starks, Pratt and Last. Official publisher search results provide the nine-module outline and Access practice description.

[P4]

Pearson: Database Concepts, 10th edition ↗

Kroenke, Auer, Vandenberg and Yoder. Current publisher educator page; the uploaded PDF is the older ninth edition, not the tenth.

[P5]

Pearson: Modern Database Management, 14th edition ↗

Hoffer, Topi, Venkataraman and Bala. Official contents and description checked October 25, 2026; explicitly covers analytics, cloud and AI.

[L1]

U.S. Bureau of Labor Statistics: Data Scientists ↗

Current Occupational Outlook Handbook read October 25, 2026. Projects 35% employment growth, 2025-2035; all occupations 3%.

[L2]

U.S. Bureau of Labor Statistics: Database Administrators and Architects ↗

Current Occupational Outlook Handbook read October 25, 2026. Architects 9%; administrators 0%; combined 4%, 2025-2035. Rounded percentages.

[L3]

World Economic Forum: Future of Jobs Report 2025 ↗

Employer expectations for 2025-2030: analytical thinking essential to about seven in ten responding companies; AI and big data lead skills expected to grow. Global employer survey, not U.S. hiring requirements.

[T1]

PGlite: What is PGlite? ↗

Official documentation: embedded PostgreSQL compiled to WebAssembly; runs in the browser, with optional local persistence. It is not itself a hosted cloud database.

[T2]

Supabase: Database overview ↗

Official documentation: each project provides a PostgreSQL database. Hosted operations, accounts, permissions and product limits require a separate teaching plan.

[T3]

Microsoft: Get Power BI Desktop ↗

Current installation documentation, checked October 25, 2026. Windows desktop authoring creates an access requirement for Mac users.

[M1]

NCES: Undergraduate Degree Fields ↗

2021-22: 375,400 business bachelor’s degrees, 19% of bachelor’s awards. Broad audience context; degree completions are not database-course enrollments.

[M2]

Cengage: Database Systems, 15th edition, U.S. product listing ↗

Pricing retrieved October 25, 2026: eBook US$71.99 through January 22, 2027 or US$98.99 through July 21, 2027; paperback US$272.95; MindTap advertised from US$149.00. Prices and access dates can change.

[C1]

Original cover on the current book website ↗

Existing animated cover URL verified in the current repository’s HomePage component. Used unchanged, with a still-image fallback for reduced motion.